GLP-1 Demand Surge Intelligence

Time-aware healthcare demand modeling with public aggregate data

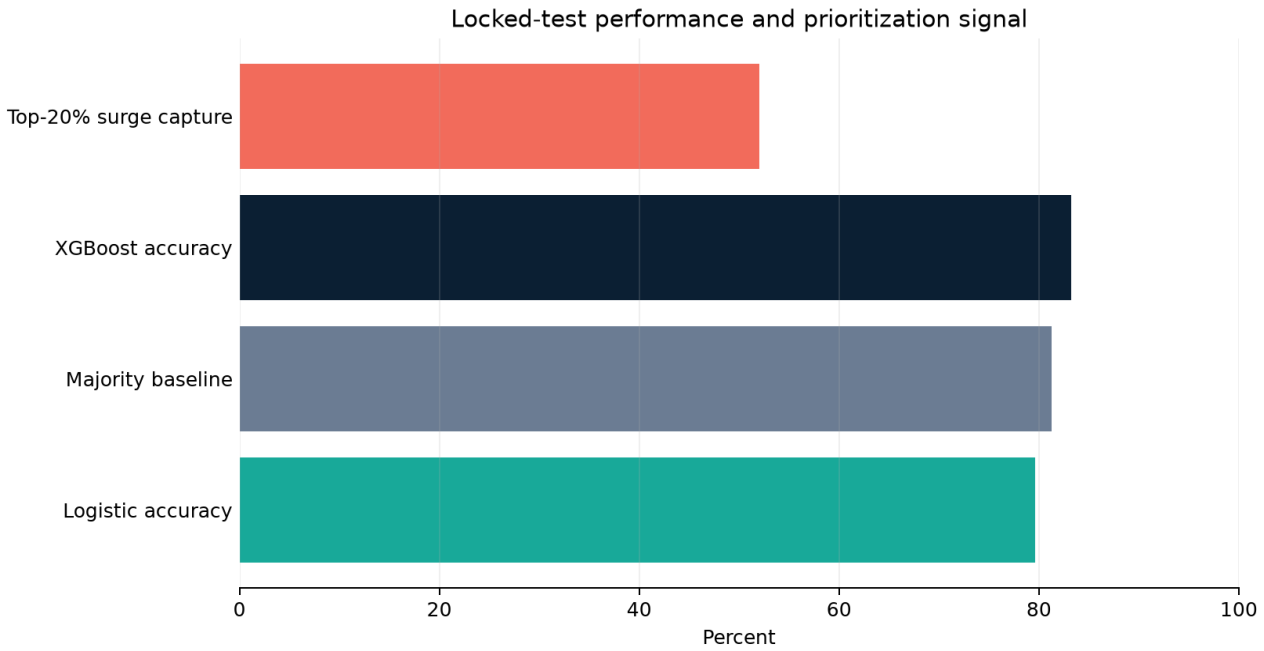
Portfolio Case Study

Executive Summary

This analysis built a chronological classification workflow from public CMS utilization and enrollment data plus FDA product metadata. The final analytic sample contained 10,272 strict-history observations. Logistic regression produced 79.6% test accuracy but did not beat the 81.3% majority baseline, while XGBoost improved accuracy to 83.2% and ROC-AUC to 0.791. The strongest decision use is monitored analyst prioritization, not automated ordering.

Decision Question

Can historical utilization and market signals identify state-product markets at risk of a next-quarter prescription-demand surge?



Methods

- Constructed state-product-quarter panels using public, aggregate administrative sources.
- Engineered lagged utilization, growth, market-era, product-age, and enrollment features using only historically available rows.
- Separated training, model-selection, and locked test periods chronologically to reduce future-data leakage.
- Compared interpretable logistic regression with a nonlinear XGBoost benchmark and evaluated ranking, calibration, and threshold behavior.

Evidence and Findings

Evidence	Observed result
Analytic sample	10,272 strict-history observations; 1,878 rows in the locked test period.
Logistic model	79.6% accuracy, 24.4% recall, and performance below the 81.3% no-surge baseline.
XGBoost benchmark	83.2% accuracy, 0.791 ROC-AUC, and 20.5% recall at the selected threshold.
Prioritization value	The highest-ranked 20% of test observations captured 52.0% of observed surges.

Limitations

- Suppression and strict-history rules emphasize observable, higher-volume state-product quarters.
- Administrative publication lag separates retrospective results from a true real-time forecasting environment.
- Aggregate data cannot identify diagnosis, indication, prescriber intent, inventory, or patient-level behavior.
- Repeated states and products constrain conventional independence assumptions and causal interpretation.

Recommended Next Actions

- Use the model as an analyst-review queue with explicit capacity and recall targets.
- Run a prospective shadow evaluation before operational use and monitor calibration and data drift.
- Retain the logistic model as an interpretability benchmark and evaluate XGBoost for ranking use cases.

Reproducibility

The companion repository contains typed Python modules, tests, dependency declarations, data-access notes, and a rendered analysis notebook. Restricted source data and the working notebook are intentionally excluded from the public repository.

Results are reported as historical analytical evidence. They should be reproduced on an authorized equivalent dataset before operational use.